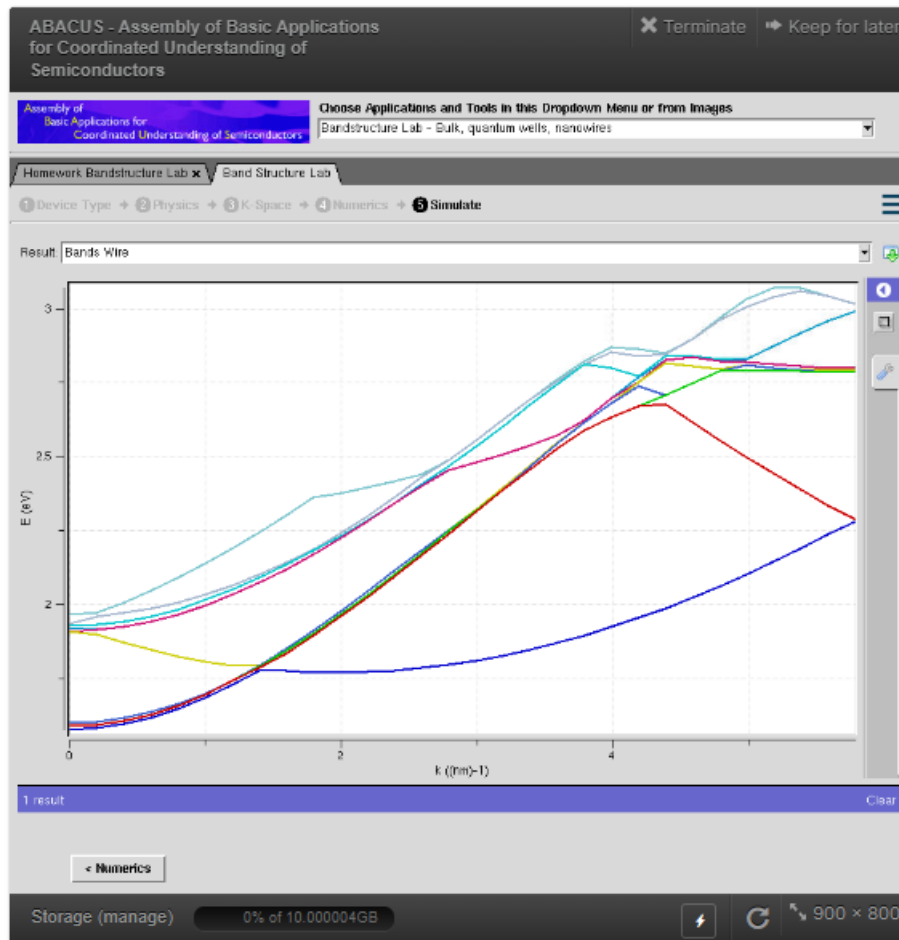
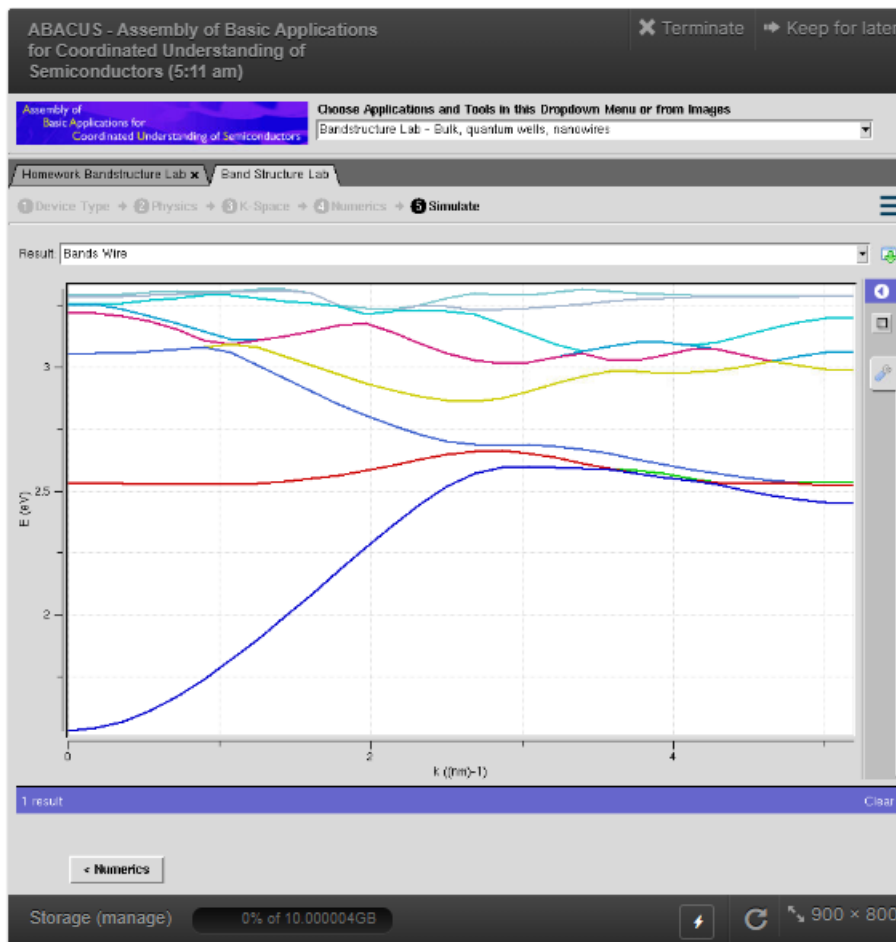


Exp. 5. Band Structure Simulation for Graphene and CNTs using NanoHub



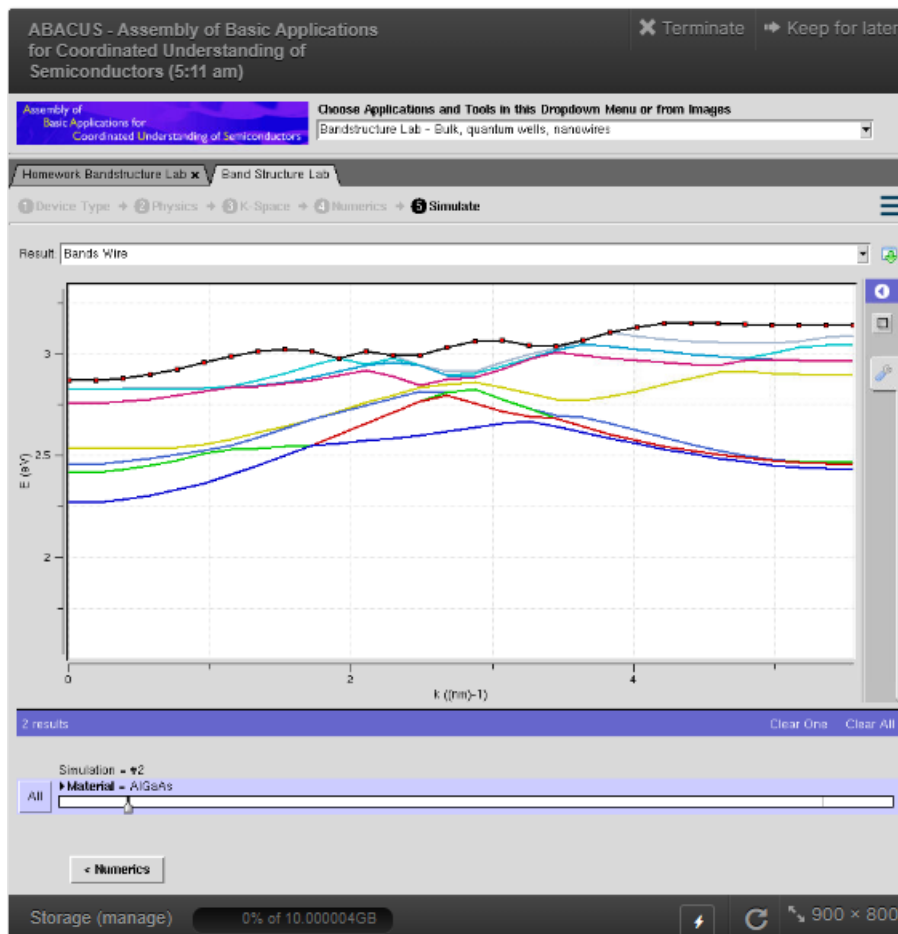
1. This image shows the **Band Structure Lab** simulation of a semiconductor using the **ABACUS** nanoelectronics tool.
2. The graph plots **Energy (E)** versus **Wave Vector (k)** to represent the electronic band structure.
3. The different colored curves indicate the **allowed energy bands** for electrons in the material.
4. The variation in the bands helps analyze **electron movement, band gap, and carrier behavior**.
5. This simulation is useful for designing and optimizing **semiconductor devices, nanowires, and quantum well structures**.

Case Study 1: Simulate the band-structure of InAs circular nanowire



1. This image shows the **Band Structure Lab** simulation of a semiconductor using the **ABACUS** simulation platform.
2. The graph represents the relationship between **Energy (E)** and the **Wave Vector (k)** for different electronic bands.
3. The multiple colored curves illustrate the **conduction and valence energy bands** of the semiconductor material.
4. The band distribution helps evaluate the **electronic properties, band gap, and charge carrier transport**.
5. This simulation is used to optimize **semiconductor materials, quantum devices, nanowires, and high-speed electronic applications**.

Case Study 2: Simulate the band-structure of AlGaAs circular nanowire



1. This image shows the **Band Structure Lab** simulation for the semiconductor material **AlGaAs** using the **ABACUS** platform.
2. The graph plots **Energy (E)** versus **Wave Vector (k)**, showing the electronic band structure of the material.
3. The colored curves represent the **valence and conduction energy bands**, while the black curve indicates the selected simulation result.
4. The band structure is used to analyze the **band gap, electron transport, and carrier dynamics** in AlGaAs.
5. This simulation helps optimize **high-speed transistors, LEDs, laser diodes, photodetectors, and other optoelectronic devices**.